

CrabTransformer Training Report

Scenario: HARD | Loss: MSE

— Data Generation Parameters —

grid_size: 50 n_years: 30 n_bootstraps: 100
mean_density: 3728.0 mean_recruit_density: 4514.0
recruitment_correlation: 0.8 lag: 5
spatial_kappa: 0.3 temporal_rho: 0.7

— Training Configuration —

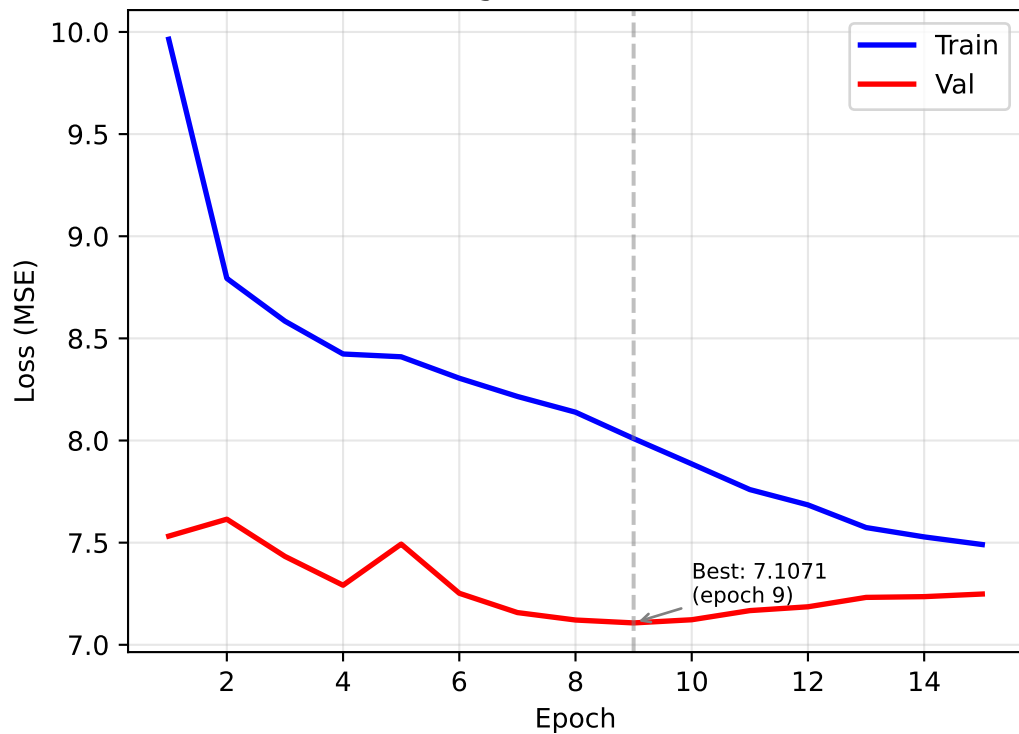
epochs_run: 15 best_val_epoch: 9
best_val_loss: 7.1071 final_train_loss: 7.4907
bias_correction (MSE): 47.4637

— Model Architecture —

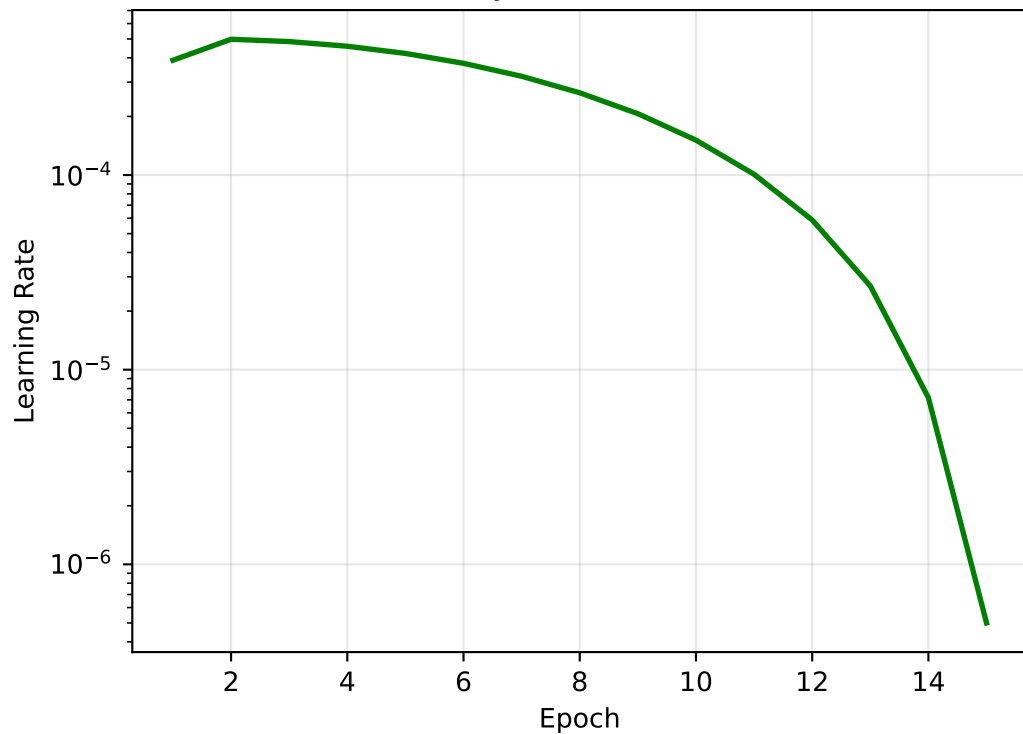
embed_dim: 128 num_heads: 8 num_layers: 6
d_ff: 512 dropout: 0.05 patch_size: 5
train/val/test years: 18/9/3

Training Curves: HARD | MSE

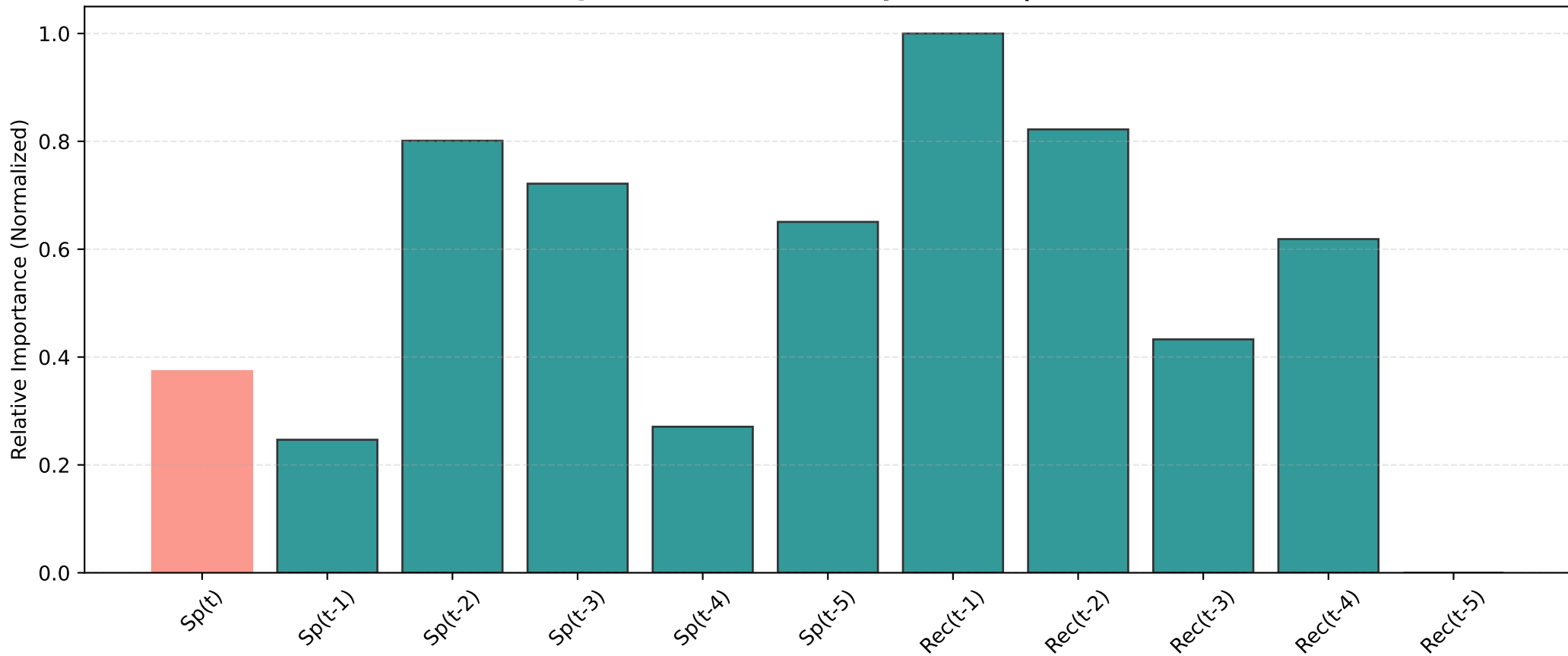
Training & Validation Loss



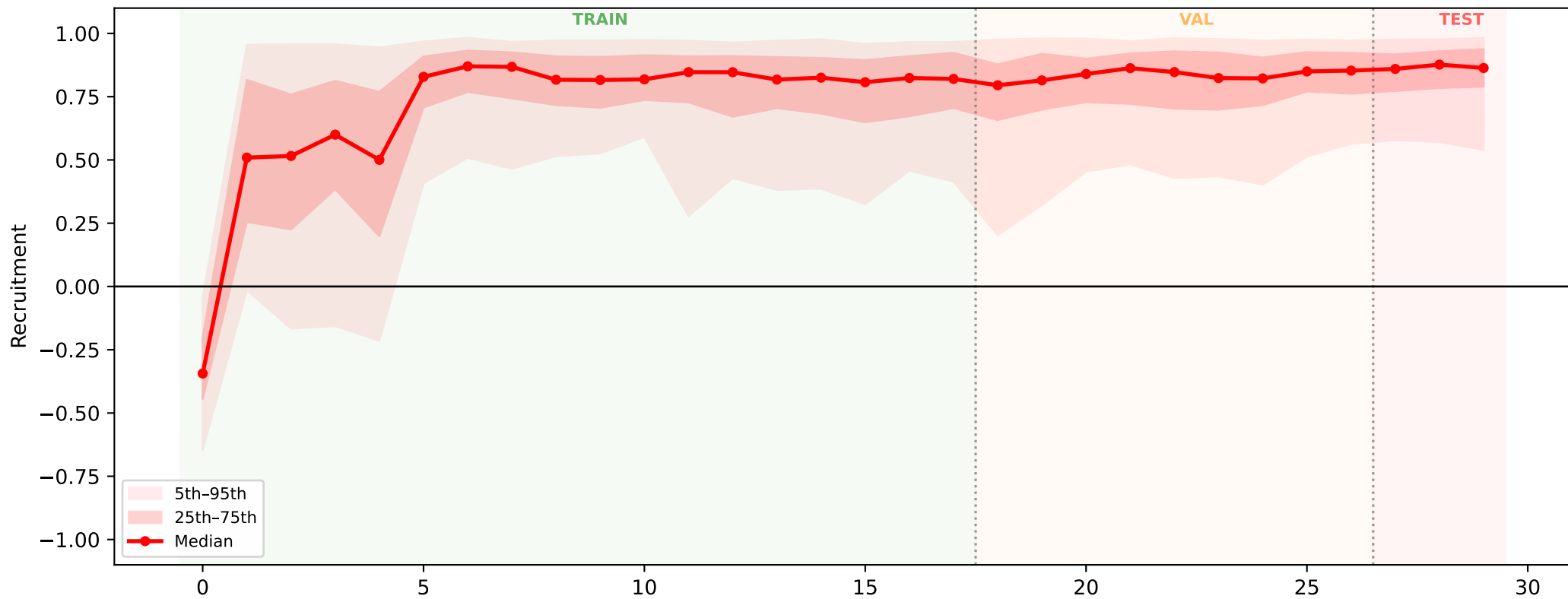
OneCycleLR Schedule



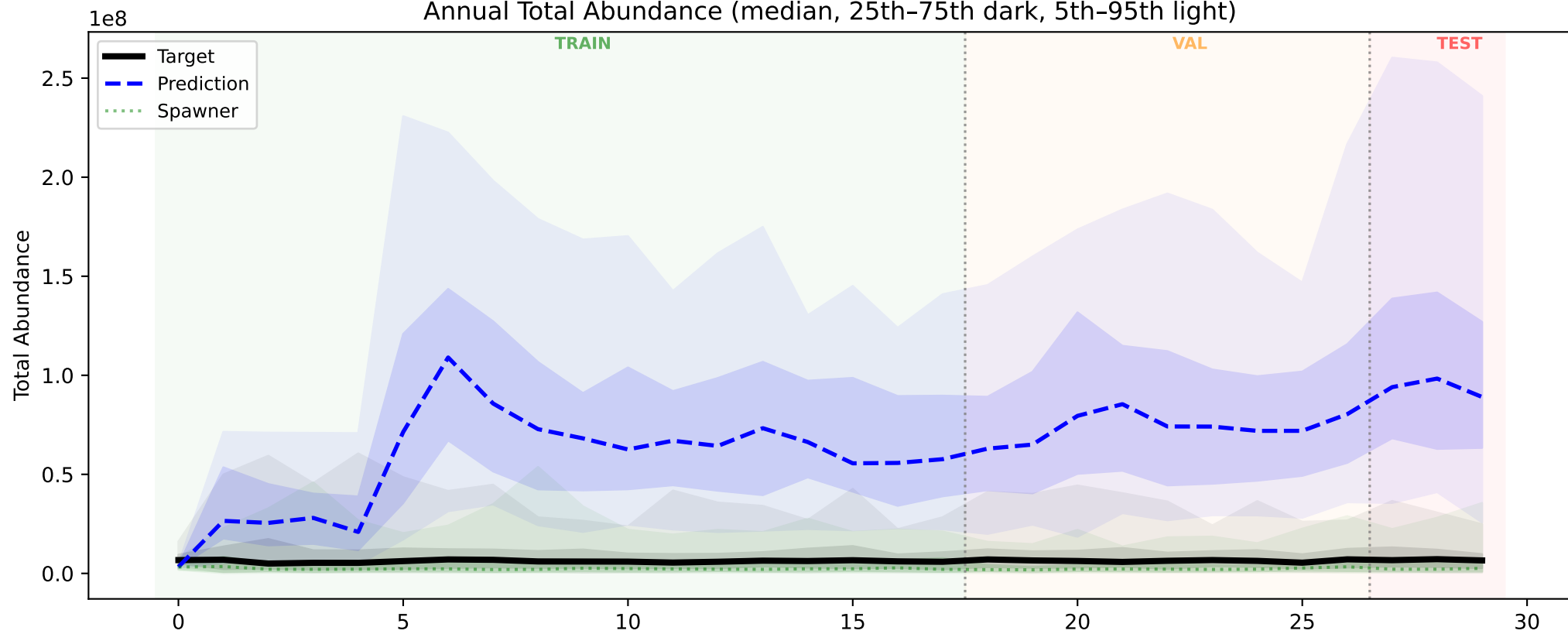
Input Channel Priority: HARD | MSE



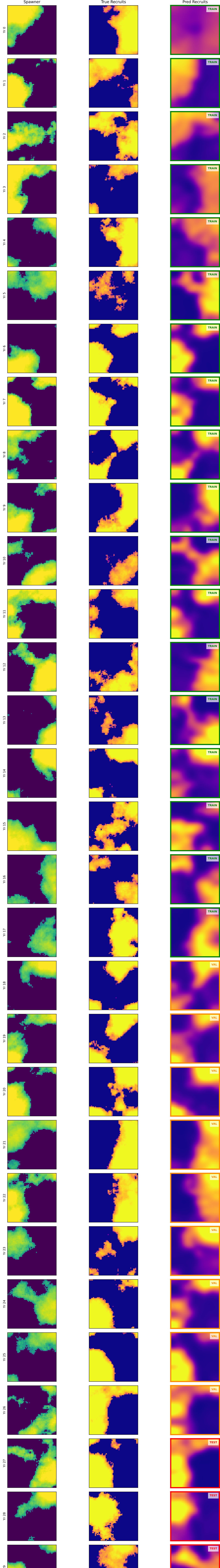
Annual Recruitment Deviations: HARD (MSE)



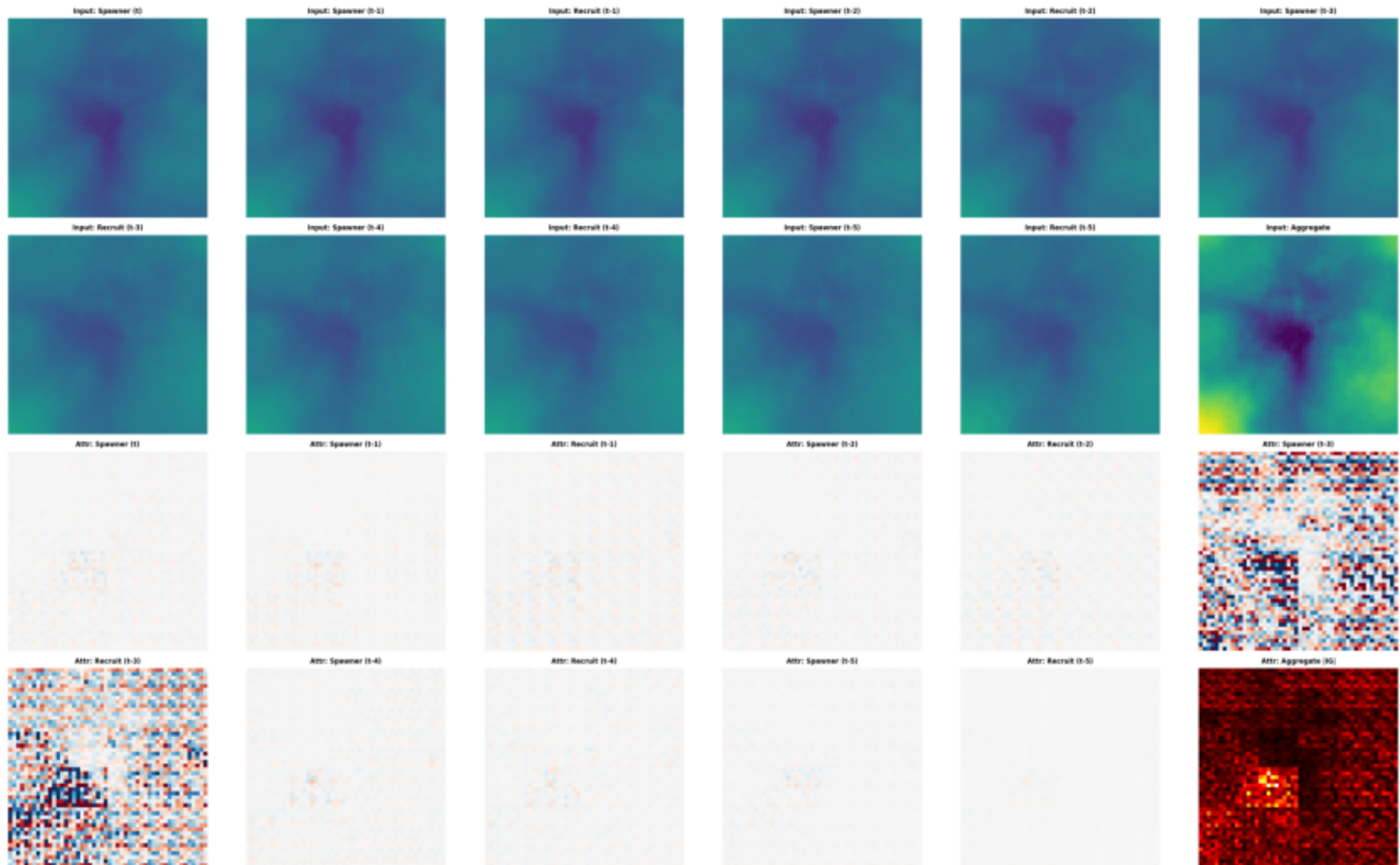
Annual Total Abundance (median, 25th-75th dark, 5th-95th light)



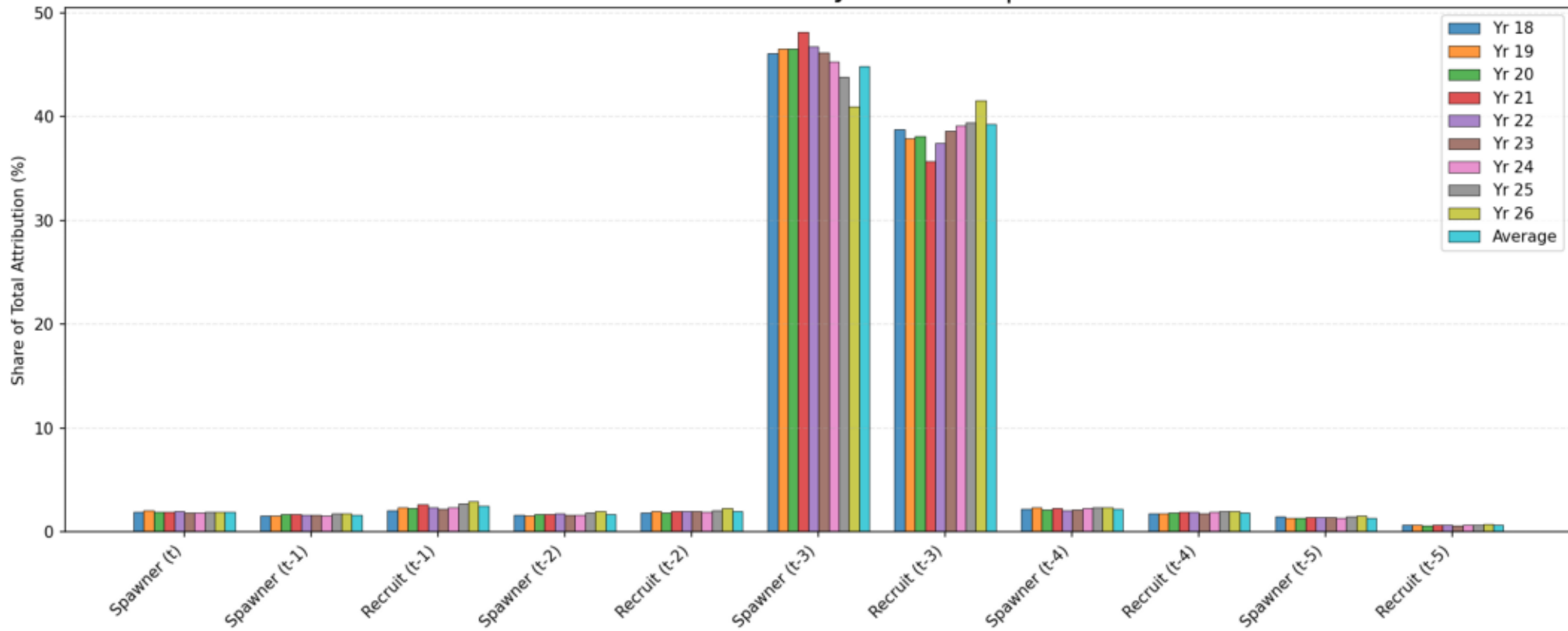
Spatial Trajectory: HARD | MSE



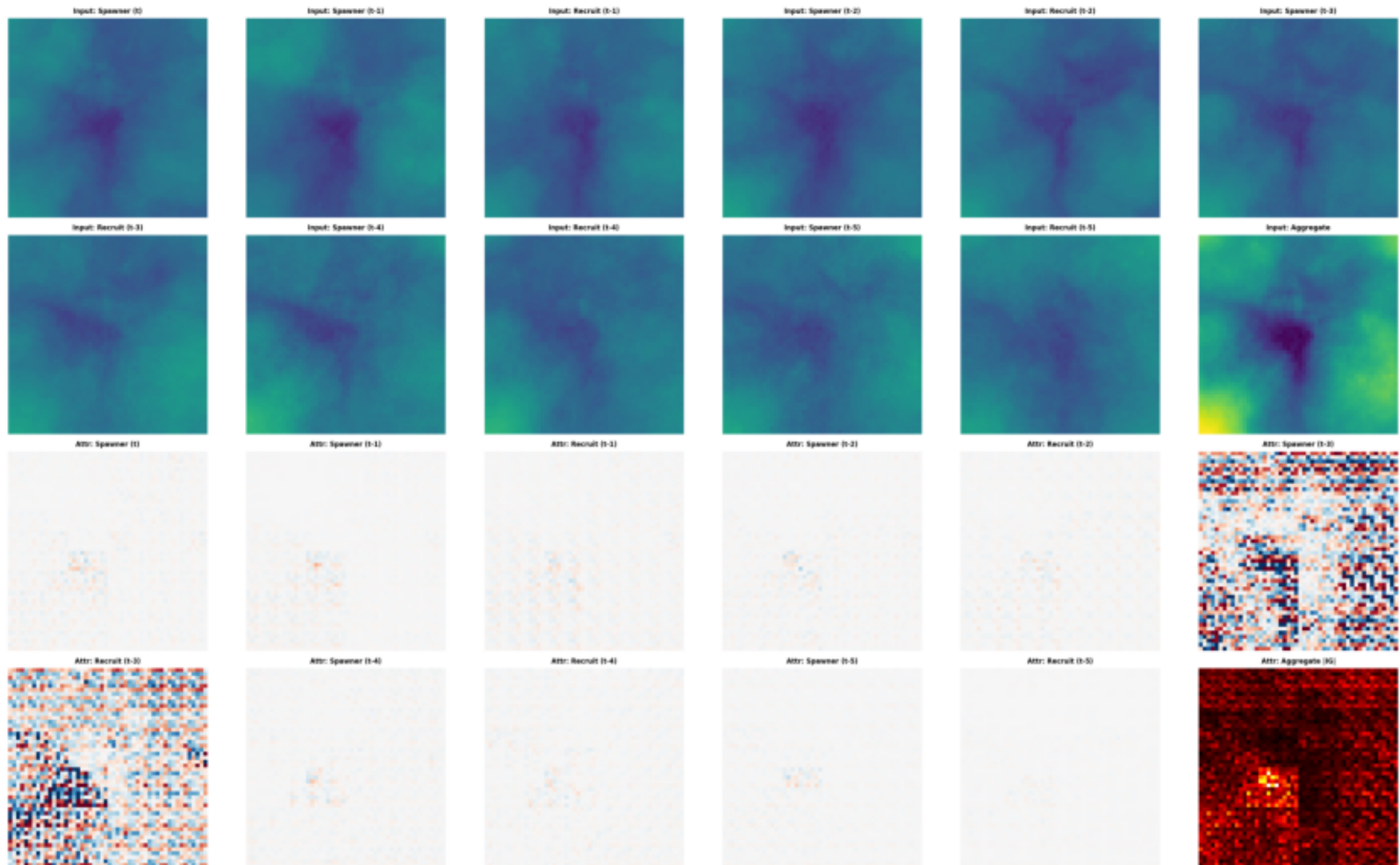
Year AVG — HARD | MSE
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



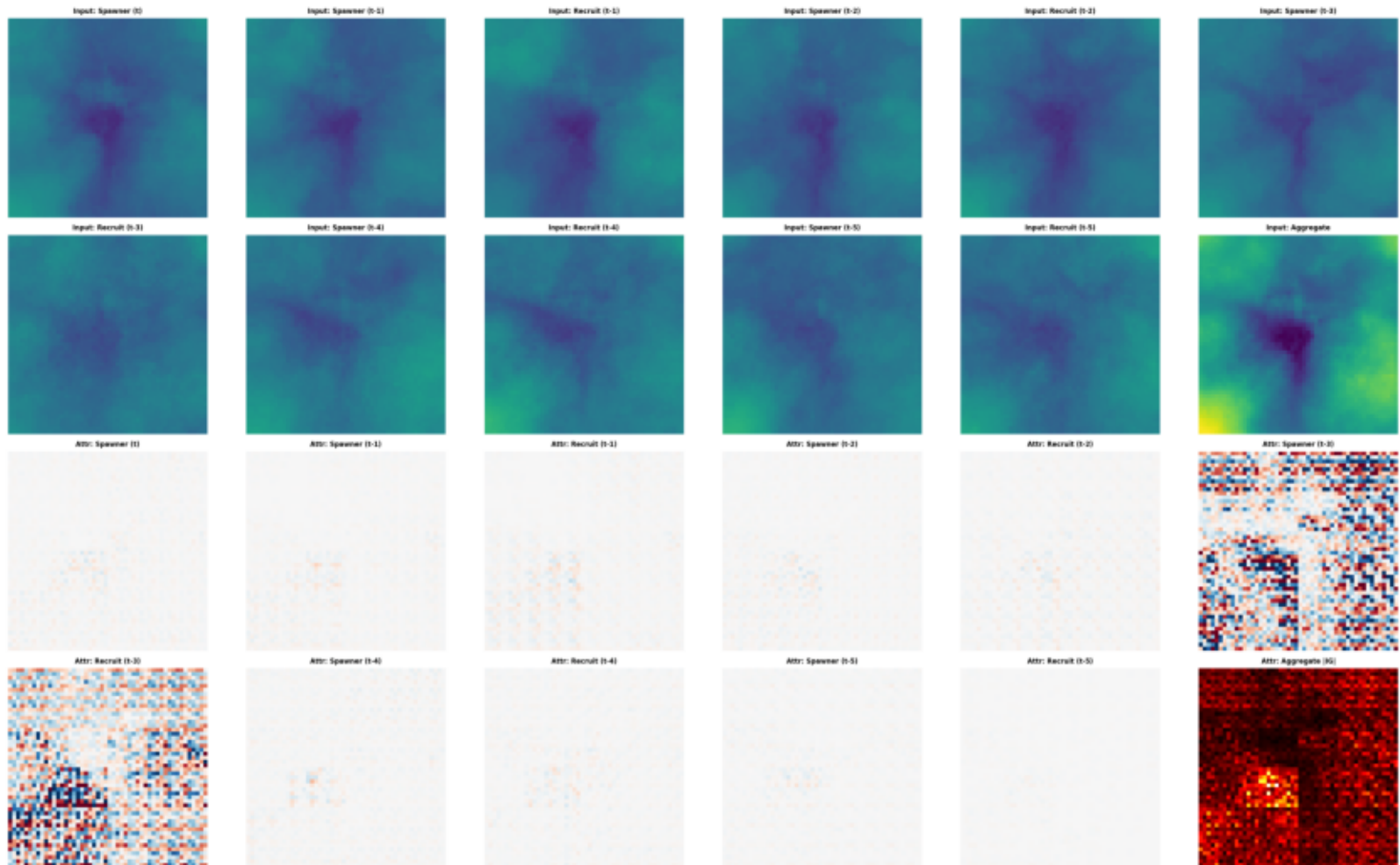
Channel Attribution by Year: HARD | MSE



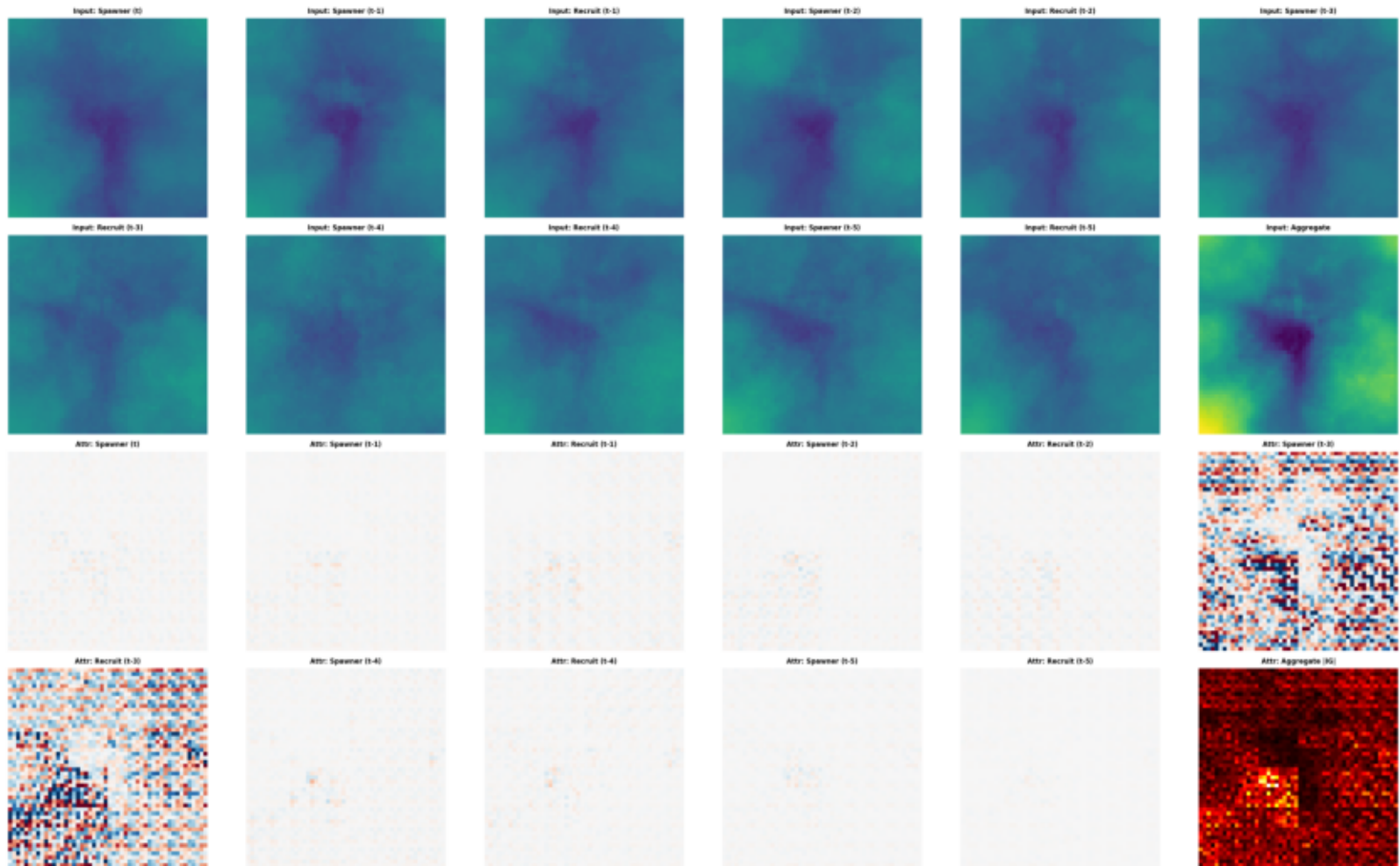
Year 22 — HARD | MSE
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



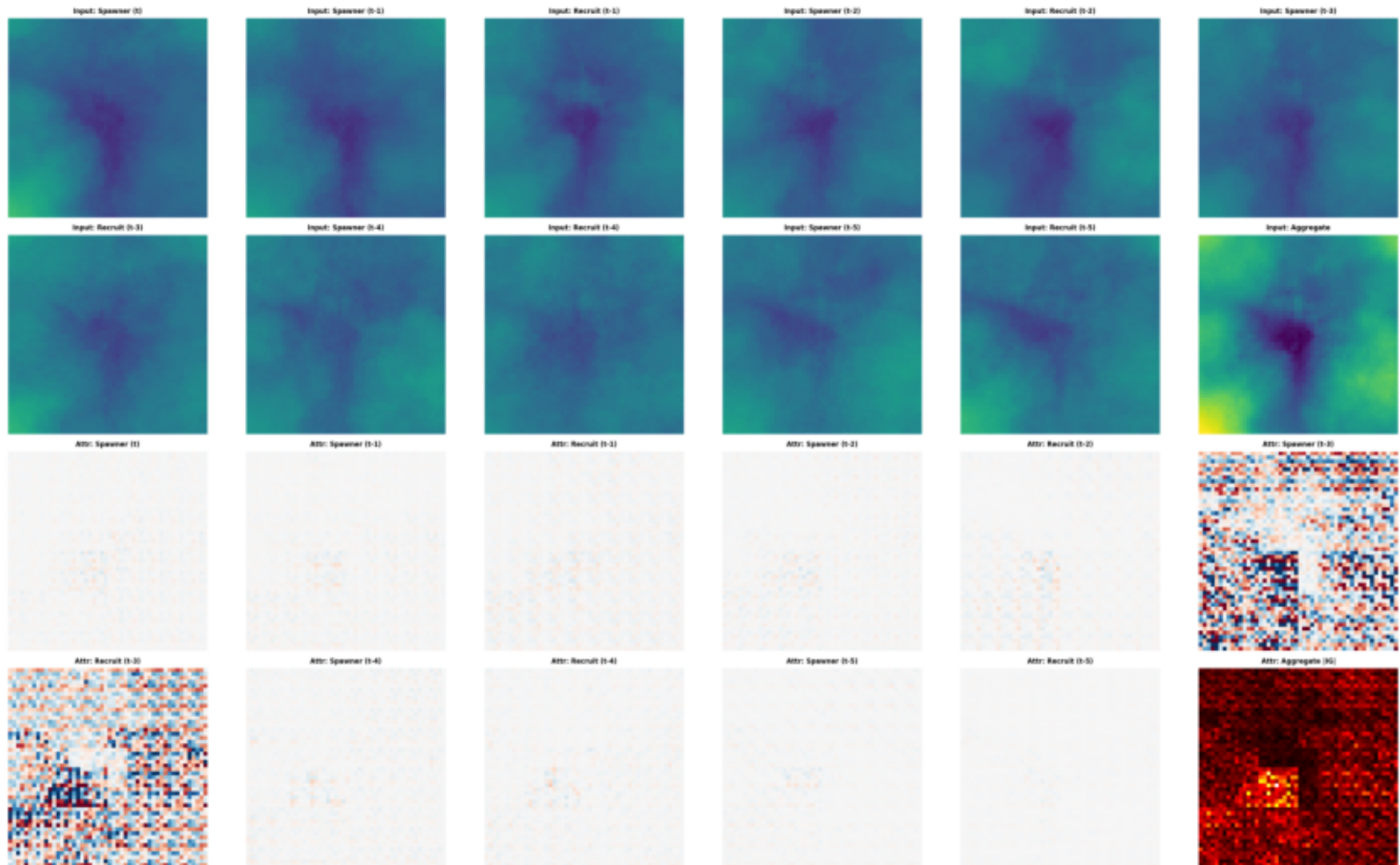
Year 23 — HARD | MSE
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



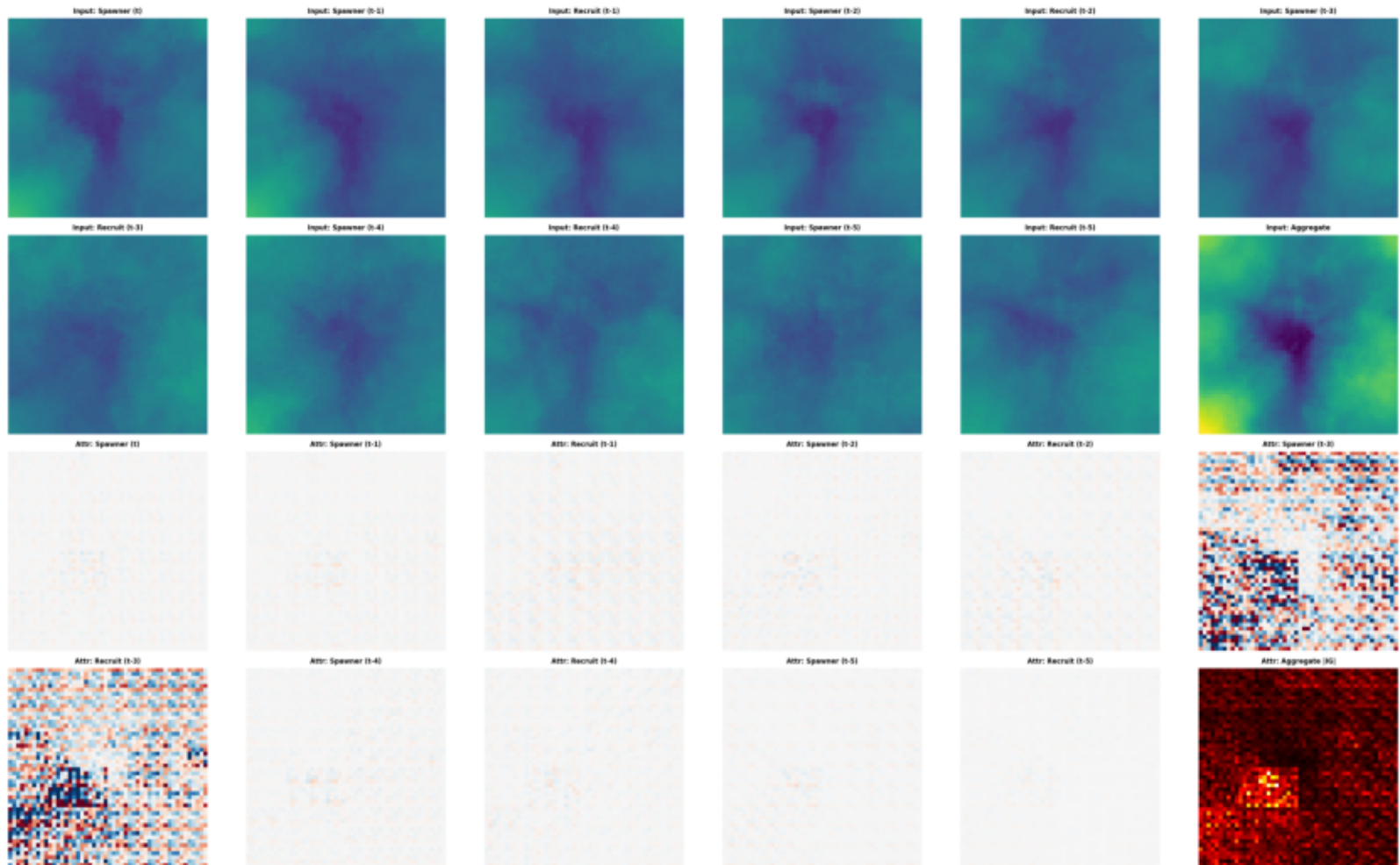
Year 24 — HARD | MSE
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 25 — HARD | MSE
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 26 — HARD | MSE
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



Summary Statistics: HARD | MSE

Phase	Pred_Mean	Tgt_Mean	Pred_SD	Pred_P50	Pred_P95	Abundance_Capture	MAE	Spearman
TRAIN	26037.5	4197.8	86871.8	384.8	152747.4	6.203	23401.1	0.657
VAL	33681.7	3921.9	99285.2	251.7	204394.2	8.588	30540.1	0.708
TEST	43977.4	3896.0	129339.8	272.3	259910.6	11.288	40699.6	0.710

Detailed Metrics (R-style): HARD | MSE

Phase	RMSE	RMSE_unbias	Bias	Bias_%	Spearman	Pearson	Pct_Error	Zero_F1	TopK_Acc	TopK_Jaccard	Quant_Err	Skill_MSE	Skill_MAE
TRAIN	29463.5	19301.6	21840.5	52.5%	0.937	0.868	544.4%	0.000	0.785	0.648	0.73	-13.935	-5.309
VAL	38550.5	24361.7	29761.0	60.2%	0.947	0.856	769.2%	0.000	0.752	0.610	0.68	-29.114	-8.284
TEST	51265.9	31876.8	40082.9	61.2%	0.948	0.839	1031.1%	0.000	0.815	0.691	0.66	-50.781	-11.857

Appendix A: Summary Metrics

APPENDIX A: Table Metric Definitions

All metrics computed on ORIGINAL DENSITY SCALE (after expm1 and bias correction for MSE). Zero threshold = 14 (smallest nonzero observed value): predictions AND observations below 14 are set to 0 before ALL metric computation. Both Page 7 and Page 8 apply this threshold.

Summary Table (Page 7)

Computed per bootstrap sample, then averaged across all samples in the phase.

Pred_Mean, Tgt_Mean

Arithmetic mean of all cell values across all bootstraps (after thresholding).

Pred_SD

Standard deviation of all cell values.

Pred_P50, Pred_P95

Median and 95th percentile of all cells.

Abundance_Capture

$\text{sum(pred)} / \text{sum(target)}$ across all cells and bootstraps. 1.0 = perfect match.

MAE

$\text{mean}(|\text{pred} - \text{target}|)$ per grid cell, then averaged across grid cells, years and, bootstraps.

Spearman

Spearman rank correlation between flattened 50x50 pred and target grids.

Computed per sample on thresholded original-scale values, then averaged.

Appendix A (cont): Detailed Metrics

APPENDIX A (cont): Detailed Metrics (Page 8)

For Page 8, metrics are computed per-year by first AVERAGING the 50x50 spatial field across all 100 bootstraps (both target and pred), then applying the zero threshold, then computing metrics on that mean field. Per-year metrics are averaged across years within each phase (TRAIN/VAL/TEST).

RMSE

$\sqrt{\text{mean}((\text{obs} - \text{pred})^2)}$ across all 2500 cells.

RMSE_unbiased

RMSE after subtracting mean bias:

$\text{corrected} = \text{pred} - \text{mean}(\text{pred} - \text{obs})$

$\text{RMSE_ub} = \sqrt{\text{mean}((\text{obs} - \text{corrected})^2)}$

Isolates spatial pattern error from overall level error.

Bias

$\text{mean}(\text{pred} - \text{obs})$. Positive = systematic overprediction.

Bias_%

$\text{bias}^2 / \text{rmse}^2 * 100$

High (>50%): main error is wrong overall level.

Low: spatial pattern errors dominate.

Spearman

Rank correlation on the thresholded, original-scale, bootstrap-averaged field.

Pearson

Linear correlation on the same field. Sensitive to magnitude unlike Spearman.

Pct_Error

$(\text{sum}(\text{pred}) - \text{sum}(\text{obs})) / \text{sum}(\text{obs}) * 100$

Zero_F1

F1 for zero-classification (threshold = 14):

TP: $\text{obs} < 14 \text{ AND } \text{pred} < 14$

FP: $\text{obs} \geq 14 \text{ AND } \text{pred} < 14$

FN: $\text{obs} < 14 \text{ AND } \text{pred} \geq 14$

$\text{Prec} = \text{TP} / (\text{TP} + \text{FP})$, $\text{Rec} = \text{TP} / (\text{TP} + \text{FN})$

$\text{F1} = 2 * \text{Prec} * \text{Rec} / (\text{Prec} + \text{Rec})$

Appendix A (cont): Ranking & Skill

APPENDIX A (cont): Detailed Metrics (Page 8)

TopK_Accuracy

Top 10% of cells by obs vs pred density:

$k = \text{ceil}(0.1 * 2500) = 250 \text{ cells}$

$\text{Acc} = |\text{obs_top250} \& \text{pred_top250}| / 250$

Measures whether the model correctly identifies spatial hotspots.

TopK_Jaccard

Jaccard index for the same top-k sets: $|\text{intersection}| / |\text{union}|$

More conservative: penalizes both missed and false hotspots.

Quantile_Mean_Error

Obs and pred independently binned into 10 quantile bins:

- Cells < 14 get bin 0

- Nonzero cells get bins 1-10 by quantile rank within nonzero values

Metric = $\text{mean}(|\text{obs_bin} - \text{pred_bin}|)$ across all 2500 cells.

0 = perfect distributional match, 3 = cells are avg 3 bins off.

NOTE: obs and pred use their OWN quantile edges, measuring distributional shape agreement, not just rank order.

Skill_MSE

1 - $\text{MSE_model} / \text{MSE_climatology}$, where climatology = $\text{mean}(\text{obs})$ everywhere.

Positive = beats the mean prediction. Negative = worse than the mean.

Primary 'is the model useful' metric.

Skill_MAE

1 - $\text{MAE_model} / \text{MAE_climatology}$. Less sensitive to outlier cells.

————— Difference between Page 7 and Page 8 —————

Both apply the same zero threshold (14) on the same original density scale.

They differ in aggregation order:

Page 7: compute metric per bootstrap, then average metrics

Page 8: average bootstraps into mean field, then compute metric

Page 8 gives model performance on the expected spatial field (less noisy).

Appendix B: Plot Methodology

APPENDIX B: Plot Methodology

— Page 2: Training Curves —

Left: per-epoch train and val loss.

MSE: loss in LOG-SPACE (on $\log(1+x)$ transformed values).

Tweedie: loss on ORIGINAL SCALE ($\expm1$ applied before loss function).

Arrow marks epoch with lowest validation loss.

Right: OneCycleLR learning rate schedule (log scale).

Ramps $\max_lr/10 \rightarrow \max_lr$ (first 10%), then cosine-anneals to $\max_lr/1000$.

— Page 3: Embedding Weight Magnitudes —

For each of 11 input channels (Spawner + 5 historical spawner-recruit pairs):

importance = $\text{mean}(|\text{weights}|)$ of channel's linear projection in PatchEmbedding.

Min-max normalized to [0, 1] for comparison.

This reflects signal AMPLIFICATION at the embedding layer – a static property of the trained weights, NOT causal importance or input-dependent attribution. Compare with the IG attribution bar chart for the gradient-based view.

— Page 4: RecDev & Abundance —

Top panel (RecDev):

$\text{RecDev} = (\text{pred_sum} - \text{target_sum}) / (\text{target_sum} + \text{pred_sum})$ per sample.

Red line = median across bootstraps. Ribbon = 5th to 95th percentile.

Near 0 = balanced. Positive = underprediction. Negative = overprediction.

Bottom panel (Abundance):

Median of grid-wide sums per year. Ribbons = 5th to 95th percentile.

Target (black): $\text{sum}(\expm1(\text{target_log}))$

Pred (blue): $\text{sum}(\expm1(\text{pred_log}) * \text{bias_correction})$

Spawner (green): $\text{sum}(\expm1(\text{spawner_log}))$

Appendix B (cont): Spatial & Attribution

APPENDIX B (cont): Plot Methodology

— Page 5: Spatial Trajectory —

Three columns per year: spawner input | true recruits | predicted recruits.
All displayed in LOG-SPACE with shared color scale (vmin=0, vmax=8).
Only the first bootstrap replicate is shown.

Prediction borders indicate phase:

Green = TRAIN, Orange = VAL, Red = TEST

— Page 6: Integrated Gradients Attribution —

Attribution computed via integrated gradients (Sundararajan et al. 2017).
For each validation year:

1. Baseline = mean training input field (the 'boring' average-year reference)
2. Interpolate from baseline to actual input in 50 steps (alpha: 0 -> 1)
3. At each step, compute gradient of mean predicted recruitment w.r.t.
the interpolated input: $d(\text{mean_pred}) / d(\text{input})$
4. Average gradients across all 50 steps
5. Scale by (actual_input - baseline) to convert rate to total contribution

Result: attribution tensor [11, 50, 50] with same shape as input.

Each value = estimated contribution of that pixel/channel to mean predicted recruitment. Averaged across bootstrap replicates per year to reveal stable spatial patterns.

Top rows: mean input channels (log-space)

Bottom rows: attribution per channel (RdBu colormap)

Red = increases recruitment prediction, Blue = decreases

Aggregate panel = $\sum(|\text{attribution}|)$ across all 11 channels

Appendix B (cont): Bars & Corrections

APPENDIX B (cont): Plot Methodology

— IG Channel Bar Chart —

Total |attribution| summed across all spatial cells for each of the 11 channels, expressed as percentage of total. Grouped by validation year plus grand average across years.

This is the gradient-based, input-dependent counterpart to the static embedding weight magnitudes on Page 3. Agreement between the two provides converging evidence for channel importance; disagreement indicates the transformer layers modulate channel usage beyond what embedding weights suggest.

— Phase Region Markers —

All time-series plots include:

- Vertical dashed lines at year 21.5 (train/val) and 26.5 (val/test)
- Shaded backgrounds: green (train), orange (val), red (test)
- Text labels at top of each region

Year indices are 0-based:

0-21 = TRAIN (22 years), 22-26 = VAL (5 years), 27-29 = TEST (3 years)

— Bias Correction (MSE only) —

$\text{pred_original} = \text{expm1}(\text{pred_log}) * \exp(\text{sigma}^2 / 2)$

where $\text{sigma}^2 = \text{var}(\text{target_log} - \text{pred_log})$ computed on the training set using the best checkpoint. Corrects Jensen's inequality: $E[\exp(X)] > \exp(E[X])$ when X has variance.

Tweedie: no correction needed – loss is computed on the original scale (expm1 is applied inside the forward pass before loss), so gradients naturally optimize for the correct conditional mean.